

Compositional Perfectionism: Moral Framework for Experiential Landscapes

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1 I. The Problem with Welfare

Consider what standard welfarist ethics asks us to measure. It takes the hedonic state of a subject—how good or bad things feel, moment by moment—as the fundamental unit of moral concern, and it evaluates lives by aggregating these states over time. The appeal is obvious: valence is a single coordinate, it admits of summation, and it appears to capture what we naively mean when we say a life is going well. We do not dispute that valence matters. We dispute that it is the *whole* of what matters, and we will argue that the conflation of hedonic state with experiential richness is the central error from which the rest of this framework departs.

To see the difficulty, imagine two lives whose mean valence is identical—the integral $\frac{1}{T} \int_0^T v(t) dt$ returns the same number for both. The welfarist calculus is, by construction, indifferent between them. Yet one may consist of a single comfortable day repeated for decades, while the other traverses many distinct modes of attention, struggle, and discovery. If your intuition registers a difference here, that intuition is tracking something the hedonic instrument cannot detect: not how the trajectory *felt* on average, but what *shape* it had.

We hasten to add that this observation has a distinguished pedigree, and we claim no credit for it. The shape-of-a-life literature established decades ago that the temporal structure of a life carries value that no hedonic sum can recover. Velleman (1991) argued that a life of ascent—early hardship resolving into late achievement—is preferable to its mirror image even when the integrals match, because narrative relations among moments are themselves bearers of value. Slote observed that we weight the temporal location of goods within a life in ways no aggregative calculus licenses. Dorsey pressed the point that whole-life evaluation is not reducible to the evaluation of its temporal parts taken severally. The divergence between summed valence and lived shape is, in other words, a settled result, and we inherit it rather than discover it. What the literature established philosophically, however, it left geometrically uncharacterized: *shape* remained a metaphor, gesturing at narrative arc or directional trend without specifying what mathematical object a life's shape is, in what space it resides, or by what instrument two shapes might be compared. The inheritance, then, comes with an unpaid debt, and it is precisely this debt that the present framework undertakes to discharge.

Our departure, then, is to give shape a structural object and a measurement program. The claim is not that lives have shape—that much is settled—but that shape can be characterized geometrically, compared across subjects, and made the basis of obligation. Concretely: we will take the moral object to be the subject-side trajectory $\tau = (S_0, S_1, \dots, S_T)$, the ordered history of states a configured subject passes through, and we will ask of that trajectory the questions one asks of any geometric object. How many in-

dependent directions does it explore (its effective dimension)? Does it admit a short description (its compressibility)? Does the landscape it traverses offer gradients, exits, and learnable structure (its navigability)? Each of these has, at least in principle, a measurable correlate, and the difference between the repeated comfortable day and the life of struggle and discovery—invisible to the hedonic integral—becomes a difference between a low-dimensional periodic orbit and a high-dimensional path that resists compression. The reader should hold us to two standards in what follows: that the geometry does genuine normative work rather than redecorating familiar judgments, and that every quantitative claim arrives with its instrument, its dataset, and its conditions of failure specified in advance.

2 II. The Moral Object

We begin with a deceptively simple question: when we ask whether a life went well, what exactly is the thing we are evaluating? Our answer is the *subjective trajectory*, which we write

$$\tau(\theta, X_{1:T}) = (S_0, S_1, \dots, S_T),$$

the ordered state history that results when an external episode $X_{1:T}$ passes through a configured subject profile θ . The notation earns its place by marking a distinction that ordinary language blurs. The same episode—the same storm, the same conversation, the same diagnosis—produces radically different trajectories in differently configured subjects, because each state S_t is generated by the subject’s own apparatus of attention, expectation, and memory acting on what arrives. The input stream is public; the trajectory is not.

Notice what follows immediately. If the moral object is the trajectory, then no inventory of external circumstances, however exhaustive, settles the evaluative question, and no instantaneous reading of the subject’s condition does either. A trajectory is extended, ordered, and shaped; its moral character lives in properties that only the whole path possesses—its dimension, its navigability, its resistance to summary. We will insist on this throughout: the unit of moral concern has temporal structure, and theories that discard that structure discard the phenomenon.

Let us be precise about what each state S_t contains, because the rival theories we mean to displace can each be read as keeping one component and discarding the rest. The hedonist keeps a single coordinate—valence—and treats the trajectory as a time series of one number. The preference theorist keeps the value function and asks only whether the world conformed to it. Both extract a scalar from a structure that is not scalar. What a state actually comprises, on our account, is the full subject-side record at that moment: the slice of the input stream the subject’s attention selected,

the expectation held against it, the prediction error registered when the two diverged, the surprise that error generated, the memory updated in consequence, the evaluation assigned, and the accumulated topology of where the subject has been. Pleasure appears in this inventory, but as one coordinate among several, not as the whole. To evaluate a life by its valence alone is to assess a symphony by its average loudness. The components interact—surprise reshapes memory, memory reshapes expectation, expectation reshapes attention—and it is precisely these interactions, unfolding in order, that give the path its moral character.

Nothing in this construction mentions neurons, and the omission is deliberate. The trajectory is defined by what a system *does*—by the functional organization that generates a subject-dependent state path—and not by the material in which that organization happens to be realized. Wherever we find temporal coherence binding the states into a genuine history, structured attention selecting among available inputs, memory that updates in response to prediction error, and an evaluative function assigning value along the way, there we find a trajectory τ , and there the framework’s claims apply. Carbon satisfies these conditions; nothing in the argument requires that only carbon can.

We anticipate the objection that this is mere permissiveness. It is, rather, a discipline that cuts in both directions. The criteria are demanding enough to exclude systems that merely transform inputs without accumulating a history—a thermostat has no topology of where it has been—yet they are stated abstractly enough that we cannot smuggle in a biological privilege we have not argued for. If a silicon system, or some architecture not yet conceived, produces ordered states bearing the full subject-side structure, then it possesses the very object we have identified as the unit of moral concern, and consistency forbids us from looking away.

3 III. The Grounding Question: Constitutive or Di-agnostic?

Before building anything further on the trajectory, we must confront the hardest question the framework faces, and we must confront it now rather than defer it to a methodological appendix where foundational embarrassments customarily go to be forgotten. The question is this: why is trajectory geometry *good*? When we praise high effective dimension, navigable gradients, and resistance to compression, are we identifying what value *is*, or merely pointing at an unusually reliable correlate of goods we already hold under older and more honorable names—learning, hope, agency, love?

The dilemma has teeth on both horns. If geometry is merely a correlate, then the older vocabularies retain normative priority, and our apparatus is instrumentation in the service of Nussbaum rather than a successor to her:

useful, perhaps, but philosophically derivative. If geometry is constitutive, then we owe an argument for that strong claim, and we must face the ugly cases it generates—a maximally high-dimensional but meaningless churning of states, set against a low-dimensional life of deep devotion. Intuition sides with the devotee; raw geometry, naively applied, sides with the churn. A framework that cannot say which of its claims would yield in such a collision, and why, has not yet earned the right to its formalism.

Our answer is a hybrid, and we ask the reader to hold its two tiers apart, because the framework’s coherence depends on the separation. The first tier is constitutive, and it concerns navigability alone. Gradients, exits, learnable structure—these are not signs that point toward open agency from outside; they are what open agency *is*, rendered in structural terms. For any subject capable of caring how its own path continues, a navigable landscape is the precondition of that caring having an object at all: there is nothing it could be to care about one’s continuation in a space where no direction differs from any other. Collapse, accordingly, does not *correlate* with the destruction of agency; it is that destruction, described geometrically. This claim we assert without apology, and we will not revise it against intuition, because intuition about agency is precisely what it articulates. The second tier is diagnostic, and it covers dimensionality, diversity, and incompressibility. These we defend more modestly: as the best available structural tracker of the plural goods—learning, love, growth, depth—that older vocabularies name directly. Tracker status entails revisability. Where the metrics and the goods they track demonstrably diverge, the metrics, not the goods, must give way.

This two-tier discipline governs everything that follows—but we must be careful about when it is invoked. The tier assignment decides genuine collisions between geometry and considered judgment; it must not be summoned to repair what is merely a misapplication of the geometry itself. Consider the devoted contemplative, the case that seems to embarrass us most. Her behavioral transcript compresses to almost nothing: the same cell, the same hours, the same prayers. But the framework’s object was never the behavioral transcript; it is the subject-side path, and the contemplative traditions describe that path’s state space with striking explicitness—ladders, mansions, stations, an architecture of interior ascent. Pointed at the right object, the theory natively scores devotion as rich. No collision occurs, and no judgment need override anything. The genuine residual case is the opposite one: the chaotic churner whose effective dimension is maximal and whose life means nothing. There, raw dimensionality and reflective judgment truly part ways, and there the diagnostic tier yields, as its tracker status requires. Our maxim, then: geometry proposes; reflective judgment disposes—but disposes rarely, reluctantly, and only after we have verified that the geometry was aimed at the trajectory rather than at its shadow.

4 IV. Landscape Navigability

We arrive now at the individual-level criterion, and we ask the reader to notice what it is *not*. The question this framework poses to a life is not “are you happy?” but “can you move?” The hedonic question interrogates a coordinate of the current state; ours interrogates the geometry of the landscape through which the trajectory τ is passing. Consider why this substitution is forced upon us. A subject reporting mild contentment may be sitting at the bottom of a basin from which no gradient ascends; a subject reporting strain may be climbing a slope rich with learnable structure. Valence, sampled at any instant, cannot distinguish these cases, yet everything of moral consequence separates them. What we require instead is a criterion stated in terms of the loss landscape itself: whether it admits patterns the subject can learn, whether directions of improvement exist and are discoverable, whether the space of available moves has genuine breadth rather than a single corridor, and whether the zone of optimal challenge—neither trivial nor crushing—is reachable from where the subject stands. We will shortly aggregate these features into a schematic quantity $Q(W)$, but the intuition precedes the formalism: a life is well-situated when its landscape is worth traversing.

Let us name the four features precisely. *Structure* requires that the landscape contain regularities at all: patterns that exist independently of the subject and are, crucially, learnable by the subject’s actual machinery. A landscape of pure noise fails this condition however energetically one explores it. *Gradients* require that improvement be locally discoverable—that from the subject’s present position there exist directions along which effort yields advance, rather than a plateau on which all moves are equivalent. *Dimensionality* requires that these directions be plural and independent: a single corridor, however steep its ascent, is a qualitatively poorer space than one admitting genuinely distinct paths, because it offers no recourse when the one path is blocked. Finally, *access to L^** requires that the zone of optimal challenge—difficulty calibrated to capacity, neither trivial nor crushing—be reachable from the subject’s position at least intermittently. We emphasize that L^* is not our invention. It names what Csikszentmihalyi identified as the flow channel, and we inherit the concept with citation rather than rebranding it; our contribution is only to locate it as one condition among four within a unified geometric criterion. Together these features specify what a traversable landscape must offer.

The payoff of this criterion is a verdict the hedonic instrument cannot return. Consider a subject whose valence reading is unobjectionable—mildly positive, stable, untroubled—but whose landscape has quietly emptied: the gradients exhausted, the corridors reduced to one, the zone of optimal challenge long since drifted out of reach. Every welfarist gauge pronounces this life fine, because by its lights nothing has gone wrong; no suffering registers,

no preference is frustrated. Yet on our criterion the life fails, and we want the reader to feel why the failure is real rather than aesthetic. What has been lost is not pleasure but *possibility*: the structural conditions under which the subject’s continuing could amount to anything more than persistence. Conversely, a life under genuine strain—negative valence, effortful ascent—may satisfy the criterion completely, provided the strain is the texture of a traversable slope. We thus arrive at the inversion that gives this section its force: navigability is orthogonal to feeling good, and where the two diverge, the geometry governs. Utilitarianism is not wrong about the comfortable stagnant life; it is *blind* to it, lacking any coordinate in which the impoverishment could appear. We have now built the coordinate, schematic though $Q(W)$ remains.

5 V. Collapsed-Attractor Suffering

We begin with a definition, and we ask the reader to hold each clause in mind separately before assembling them. Collapsed-attractor suffering is the *conjunction* of four conditions: sustained negative valence, near-minimal effective dimension d_{eff} , absent exit gradients, and a rigidified transition skeleton. Note that no single conjunct suffices. Negative valence alone describes ordinary pain; low dimensionality alone describes mere monotony; absent gradients alone might characterize a plateau one has not yet tried to leave. It is their co-occurrence that defines the regime we are isolating: the subject feels terrible, has almost no independent directions of experiential variation remaining, perceives no direction in which movement would improve matters, and traverses the same few state transitions in a loop that has hardened against perturbation. Geometrically, the trajectory has degenerated from a path into a fixed point—or a tight, repetitive orbit—at the bottom of a basin whose walls admit no ascent. Because each conjunct is itself graded, the category names a region of state space with a structured interior rather than a binary classification, and the moral force we will attach to it scales with depth in that region. We emphasize the conjunctive structure now because everything downstream—the lexicographic priority, the triage ordering, the retribution fork—depends on it.

The reader may suspect that this description is a metaphor reaching for scientific respectability. We want to insist on the opposite reading: it is a convergence with science already underway. Computational psychiatry now models severe depression explicitly as an attractor state—a region of the dynamical landscape into which the system settles and from which perturbations fail to dislodge it—and models suicidality through the construct of entrapment, the perceived absence of any exit. We did not invent this signature; the clinical literature describes it in nearly the same vocabulary we have been developing. What the framework contributes is not the de-

scription but its normative elevation: we take the descriptive signature the science has identified and make it the *deontic floor* of a moral theory. The implication deserves to be stated baldly, because it is contestable and we do not wish to smuggle it past the reader. On this view, hopelessness is *constitutive* of the worst wrong, not merely an aggravating factor layered atop intensity. A brief agony, however severe, occupies a different moral category from a mild misery with no exits, because the former retains the geometric structure—gradients, dimensions, a credible elsewhere—that the latter has lost. Intensity hurts; exitlessness destroys.

Let us now make precise a structural feature we have so far only gestured at: collapse is graded, not binary. Each of the four conjuncts admits of degree—valence can be more or less negative, d_{eff} can contract partially or nearly to zero, the nearest credible exit can be distant or vanishing, recurrence can be loose or rigid—and so the category inherits a severity ordering along these four axes. The conjunction defines the region; the severity ordering structures its interior. Moral force, accordingly, scales continuously: a strong presumption against permitting the condition near the region’s boundary, hardening into an inviolable constraint in its depths.

We draw attention to this architecture because it is what immunizes the category against sorites abuse. A binary threshold invites gerrymandering: an adversary redescribes a case as falling *just* outside the line and claims the constraint does not apply. But where moral weight varies smoothly with depth, redescription at the margin purchases almost nothing—shifting a case slightly toward the boundary shifts its claim on us only slightly. There is no cliff to argue a suffering being off of. The gradient does the normative work that a bright line could only pretend to do, and it does so without surrendering the absoluteness that governs the interior.

6 VI. The Lower Bound

We state the framework’s first commitment as a strict ordering rather than a weight. The lower bound holds *lexicographic* priority over every other consideration the theory recognizes: no being may be left trapped in collapsed-attractor suffering, and this prohibition is not one term among many in an aggregate to be maximized. Let us be precise about what lexicographic priority means here, since the word carries real structural content. A lexicographic ordering, like alphabetical order, never permits a deficit in the first coordinate to be repaid by surpluses in later ones—no quantity of excellence at position two outranks a single failure at position one. So it is with the trajectory landscape: the claims of a being whose state path has degenerated into the collapse signature—negative valence, contracted dimension, absent exits, rigidified recurrence—are settled *before* any question of compositional value is even admissible. We do not weigh the trapped trajectory against the

beautiful ensemble; we refuse the comparison as a category error. Why insist on a firewall rather than a steep exchange rate? Because, as we argued in Section III, collapse is not merely a very bad state to be discounted heavily. It is the structural destruction of open agency itself.

The practical grammar of the constraint is therefore one of *sequence*: remediation precedes composition, always and everywhere. A collapsed attractor is not a dark passage in otherwise interesting music—a dissonance that resolution will redeem, a minor key that deepens the whole. It is a stuck needle on a damaged record: a region of the total experiential landscape where nothing moves, nothing develops, and no surrounding richness reaches. The dead zone subtracts from the composition in a way no addition elsewhere can offset, because what has been lost is not a quantity of value but a *subject* of value, a vantage from which the ensemble could matter at all. Three obligations follow in order. First, identification: we must build the instruments and institutions that find beings in collapsed states, since the signature is often invisible to hedonic and behavioral measures alike. Second, restoration: intervention aims not at comfort but at navigability—reopened gradients, recovered dimension, reachable exits. Third, structural access: we must dismantle the systemic conditions—confinement regimes, engineered scarcity, industrial cruelty—that manufacture collapse at scale, lest remediation become an endless bailing of a deliberately flooded vessel. Only when these duties are discharged does compositional ambition acquire standing to speak.

Consider, finally, what the alternative would license. Suppose we permitted even a single exchange: one being’s navigability sacrificed because the resulting ensemble scored higher on dimensionality, diversity, or incompressibility. The moment that trade clears, something in the grammar of the theory has shifted. The trapped being no longer figures as a *subject* whose trajectory the composition exists to serve; she figures as a *texture*, a patch of material whose contribution to the aggregate pattern is what confers her standing. We would have converted a conscious being into an input to an aesthetic function—and once consciousness is an input, nothing in the optimization prevents it from being a cheap one. This is the conversion the lower bound exists to block, and it blocks it categorically rather than by pricing it steeply, because a steep price is still a price, and a price invites a buyer. Notice that the prohibition asks nothing of compositional beauty except patience: the ensemble may be as rich as ingenuity allows, *after* every trajectory within it can move. We do not thereby diminish the compositional project; we secure its intelligibility. A landscape is beautiful only to those who can traverse it, and a beauty purchased by trapping its own witnesses has no one left for whom it is beautiful at all.

7 VII. Triage Within the Lower Bound

Let us begin by acknowledging what the lexical structure, as stated so far, cannot do. A constraint of the form “remediate all collapsed attractors before pursuing compositional ambition” is perfectly action-guiding in a world containing exactly one trapped being: we find them, we restore navigability, we proceed. But our world is not that world. At any given moment there are billions of trajectories in or near the collapse region—in cells, in famines, in chronic pain wards, in the rigidified loops of severe depression—and our remediation capacity is finite, fragmented, and slow. A priority class with billions of simultaneous claimants and no internal ordering is not a moral theory; it is a paralysis condition wearing the costume of one. Critics of lexicographic structures have long pressed exactly this point, and we concede its force. What we do not concede is that the firewall must therefore come down. The correct response is not to soften the priority of P_1 over P_3 —that boundary remains absolute—but to articulate structure *within* P_1 itself: principled grounds for deciding, among beings we cannot all reach at once, whom we reach first. The framework owes this account, and we now provide it.

The ordering we propose rests on four factors, taken in rough priority. The first is *severity*: depth within the collapse region itself, assessed along the graded conjuncts of Section VI—how negative the valence, how contracted the dimensionality, how distant the nearest credible exit, how rigid the recurrence. Deeper claims press harder. The second is *tractability*: the steepness of the restoration gradient, measured as navigability recovered per unit of intervention. Here we must be candid about an uncomfortable implication: a shallow collapse with a cheap, known exit may rightly outrank a profound one for which no exit is currently understood—though the latter thereby generates a standing research obligation rather than a license to look away. The third is *duration risk*: the probability that the attractor deepens or hardens into irreversibility if left unaddressed. At equal severity, a compounding collapse outranks a stable one, because delay is itself an intervention with a sign. The fourth is *scale with shared cause*: the number of trajectories held in the same engineered attractor. A confinement system, a famine, an industrial process—these admit structural remedies, and where causes are shared, each intervention multiplies its restorative yield across every trajectory the common mechanism holds.

Taken together, these four factors convert the firewall from a paralysis condition into something more humble and more useful: a worklist with a fence around it. We do not pretend that triage among P_1 claims is anything other than tragic. When two beings are trapped and we can reach only one, the one we do not reach has been wronged by the world, and our ordering does not dissolve that wrong; it merely ensures that our response to it is principled rather than arbitrary. But tragic is not the same as undecidable.

The directive that emerges is compact: restore the most navigability to the most trapped, fastest, attending first to collapses that compound and to attractors whose shared causes admit shared remedies.

What remains absolutely forbidden is a single, precisely specified trade: the exchange of any P_1 claim against any P_3 ambition. No compositional achievement—no flourishing of dimensionality elsewhere, no beauty in the aggregate landscape—may purchase delay in reaching a being whose trajectory has collapsed. Trades *within* P_1 under genuine scarcity are unavoidable and now ordered; trades *across* the firewall remain blocked without exception. Triage, on this account, is not a betrayal of the lexical structure. It is what the lexical structure looks like when it meets a finite world honestly.

8 VIII. Compositional Beauty

Suppose the lower bound is secured: no being remains trapped in a collapsed attractor, and navigability holds across the population. What then? A moral framework that answered only “prevent the worst” would be a theory of floors with no account of ceilings. We therefore advance a positive directive: conditional on Priorities 1 and 2, optimize for the *beauty of the total experiential landscape* across all conscious beings.

Let us be careful with the word *beauty*, since we mean something structural rather than decorative. Consider an analogy we will lean on throughout: a piece of music is not made good by maximizing its average loudness, nor a painting by maximizing total brightness. What we admire in a composition is the organized relationship among its parts — the variety of voices, the independence of their movements, the resistance of the whole to summary. We propose that the ensemble of conscious trajectories admits the same mode of evaluation. The moral object at this tier is not any individual’s welfare, already protected below, but the *shape of the collection*: how richly the population of trajectories, taken together, fills experiential state space.

This is the diagnostic tier of the theory — held firmly, but revisably. Its content decomposes into four measurable components.

The first component is population-level dimensionality, written $d_{\text{eff}}^{\text{pop}}$: the number of independent directions explored by the ensemble of trajectories taken as a whole. We emphasize *independent*. A billion beings tracing the same arc through attentional state space contribute, collectively, scarcely more dimensionality than one; the quantity rises only when beings occupy genuinely distinct regions, moving along axes the others do not. Headcount, on this measure, is explicitly not the variable of interest — a point to which we return when we consider duplication.

The second component is diversity, which we operationalize as the average pairwise distance between trajectory shapes. Where dimensionality asks

how many directions the ensemble spans, diversity asks how far apart its members actually sit: whether many experiential forms coexist on their own terms, or whether the population has been pressed, gently or otherwise, into a single template.

The third component is aggregate incompressibility: how much of the ensemble resists summary. Here we must be precise about what we are *not* valuing. A population of random walkers is maximally incompressible and morally uninteresting; static is not symphony. What we seek is organized irreducibility — structure that cannot be shortened, rather than disorder that cannot be described.

We flag the open problem honestly rather than gesture past it: the distinction between organized irreducibility and mere randomness is precisely the *sophistication problem*, and it is unsolved in general. Bennett’s logical depth attempts to capture it as the computational time required to generate an object from its shortest description; Koppel’s sophistication attempts it as the size of the structural component of that description, separated from incidental noise. Neither yields a tractable, uniquely privileged measure, and we will not pretend the theoretical situation is better than it is.

What the framework actually relies upon is weaker and, we believe, defensible: the practical, windowed, finite-resource version of the distinction. Over bounded windows, with bounded compressors, one can ask whether a trajectory ensemble shortens under summary the way structured signals do or fails to shorten the way static does — and these behaviors are empirically separable even where the limiting theory is not. The wager of the third priority is that this practical separation is benign: that no morally significant case turns on the asymptotic subtleties the windowed version ignores.

Let us be explicit about the epistemic status of this move. It is an IOU, stated as one, and the diagnostic tier inherits its terms.

9 IX. The Lexicographic Structure and Its Thresholds

We can now state the architecture in full. The framework orders its commitments lexicographically into three priorities. *Priority 1*: no being trapped in collapsed-attractor suffering — the conjunction of sustained negative valence, near-minimal effective dimension, absent exit gradients, and rigidified recurrence. *Priority 2*: landscape navigability for all — every subject’s loss landscape must satisfy $Q(W) \geq Q_{\min}$, with learnable structure, available gradients, multiple independent directions, and at least intermittent access to L^* . *Priority 3*: compositional beauty — the dimensionality, diversity, incompressibility, and flow prevalence of the total experiential landscape.

The ordering is strict in a precise sense: claims at a higher priority are not weighed against claims at a lower one. No quantity of compositional

elegance purchases the right to leave a single trajectory in collapse; no degree of population-level richness justifies rendering an individual landscape untraversable. The grounding asymmetry of Section III explains why this is not arbitrary stipulation. The first two priorities rest on constitutive ground — collapse *is* the destruction of open agency, navigability *is* its structural form — while the third rests on diagnostic ground and is accordingly corrigible. A lexical ordering between constitutive preconditions and diagnostic ambitions is not a weighting choice; it reflects a difference in kind.

Lexical orderings inherit a known brittleness, and we should be candid about the inheritance. Rawls’s strict priority of liberty drew four decades of threshold objections — what happens at the margin, where a hair’s-breadth reclassification flips an absolute constraint on or off? — and we claim no immunity from the family of worries. Our answer is threshold deontology with graded entry. Recall from Section V that each conjunct of the collapse signature is graded; the P1 region therefore has a vague *boundary* but a non-vague *interior*, and the moral force of the constraint scales continuously as a case moves from one toward the other. Near the boundary, P1 functions as a strong presumption — weighty, but in principle answerable. Deep in the interior, where valence is profoundly negative, dimension near-minimal, and no credible exit remains, the constraint is inviolable. Notice what this architecture blocks, in both directions. The redescription game — nudging a case just across the threshold to escape the constraint — gains almost nothing, because near-threshold cases already carry near-absolute weight; the gradient is steep precisely where gaming would occur. And conversely, the constraint cannot be evaded by disputing decimal places, since its binding force never depended on a sharp line to begin with.

The strictness of the ordering should not be mistaken for an opposition among the priorities in practice. Lexical structure governs how claims are *adjudicated* when they conflict; it says nothing about how often they conflict, and the empirical answer is: rarely. Consider the interventions that P1 actually demands — housing for the unsheltered, treatment for chronic pain, psychiatric care for the entrapped, protection from violence. Each of these, in restoring a collapsed trajectory, simultaneously re-opens a navigable landscape, satisfying P2, and returns a distinctive experiential form to the population ensemble, raising $d_{\text{eff}}^{\text{pop}}$ and serving P3. The being lifted from the pit does not merely cease to suffer; she resumes exploring regions of attentional state space that her collapse had sealed, and the composition is richer for her resumption. This convergence is not a happy accident but a structural consequence of the framework’s geometry: collapse is dimensional contraction, remediation is dimensional restoration, and compositional beauty is dimensional abundance. The same gradient points all three ways. The lexicographic firewall, then, is best understood as insurance against the rare pathological case — the engineered hell that beautifies the whole — not as a description of ordinary moral life, where rescuing the trapped, open-

ing landscapes, and enriching the composition are most often one act under three descriptions.

10 X. The Retribution Fork

Retributive justice poses a test that we decline to pretend we have passed. Up to this point, the lower bound has been stated as though it attached unconditionally to conscious trajectories; the wrongdoer forces the question of whether that attachment survives deliberate, severe violation of others' landscapes. Consider what is actually being asked. May a community that confines a torturer also engineer his collapse—strip the gradients, seal the exits, contract the dimension—as a matter of desert? Our intuitions pull in both directions, and we have found, after sustained effort, that the framework's machinery does not adjudicate between them. We therefore present the fork as a fork. Two readings of the lower bound are consistent with everything argued so far: they agree on the diagnostic vocabulary, agree on what the collapse signature *is*, and diverge only on the scope of the beings to whom the prohibition applies. We regard this honesty as a feature rather than an embarrassment. A theory that resolved every classical standoff by fiat would deserve suspicion; one that locates precisely *where* the disagreement lives, and shows how much normative territory the disputants nonetheless share, has done genuine work. What follows states each reading in its strongest form, then identifies the floor beneath both.

The first reading holds that the lower bound attaches to consciousness as such. On this view, the prohibition on engineered collapse tracks nothing about innocence, reciprocity, or standing within a moral community; it tracks the bare fact of being a subject with a trajectory. Containment, accountability, and exclusion remain available to the community—walls are permitted—but the deliberate stripping of gradients is not, for any being whatsoever. The supporting argument is partly compositional: an engineered hell is one more dead zone in the total landscape, and the satisfactions of retribution cultivate narrow attentional states—fixation, the rehearsal of cruelty—in those who administer and witness it. Humane containment, on this reading, is a constraint internal to justice rather than mercy appended afterward.

The second reading holds that protections are sustained by a community of mutually vulnerable beings, and that systematic violation of others' landscapes forfeits the violator's full claims against restriction. Reach, power, privacy, and spontaneity may be reduced; restriction of the landscape is permitted. Yet even here, forfeiture is not metaphysical erasure: deprivation is not destruction, and the minimal standing of a conscious trajectory survives.

We note candidly that this is the familiar standoff between dignity absolutism and Morris-style forfeiture theory, inherited rather than discovered.

What the framework contributes, then, is not a resolution of the fork but a floor beneath it—and the floor has structural content that the older debate lacked. Both readings, whatever their disagreement about the scope of forfeiture, converge on a single prohibition: no one may deliberately engineer the collapse signature in another conscious being. The conjunction is fully specified—sustained negative valence, contracted dimension, sealed exits, rigidified recurrence—and so the prohibition is not a vague appeal to dignity but a determinate constraint on what may be done to a trajectory. Walls, yes; hells, no. Notice what this precision purchases. The classical standoff could agree that “cruelty” was forbidden while disputing endlessly what counted as cruel, because cruelty was characterized hedonically or rhetorically rather than structurally. Our vocabulary specifies what a hell *is*: a state space deliberately reduced to a point or a closed orbit, with the gradients stripped and the exits welded shut. A regime of confinement can now be audited against that signature directly—does it preserve learnable structure, reachable improvement, some access to optimal challenge?—rather than against contested intuitions about desert. The fork remains open above the floor. But the floor itself is now measurable, and that, we submit, is progress the inherited debate could not make on its own terms.

11 XI. Demandingness and the Two Postures

Let us begin with the concession, because it is genuine and it is large. Once a collapsed or near-collapsed trajectory enters one’s moral field—through knowledge, proximity, institutional participation, causal contribution, or membership in a scheme capable of repairing the damage—the claim it makes is not optional, not discretionary, and not a matter of personal moral style. A determinate encounter with the collapse signature activates a constraint that must be answered, in one’s proper share, before compositional projects can resume their ordinary standing. On this point the effective altruists are simply correct, and we say so without reservation.

We also owe their position its strongest form, not a caricature. The movement as it actually exists endorses *bounded* commitments: a tithe rather than total surrender, a career chosen for impact rather than a life dissolved into perpetual calculation, a cause adopted and pursued rather than a continuous re-auditing of every attachment. Its founders have themselves written against totalizing demandingness, and the median practitioner keeps her friendships, her crafts, and her unjustifiable enthusiasms. The picture of the effective altruist as a walking optimization loop is a strawman, and we decline to argue with it. What remains worth contesting is something more precise.

Our target is the limit case and, more importantly, the gravitational pull toward it: the thesis that all goods occupy a single allocative surface on which

every hour, every attachment, and every project must continuously justify itself against the global optimum. Against this thesis the inherited objections retain their force, and we deploy them by name. Williams’s integrity objection: the agent required to regard her commitments as one input among many to an impartial calculator has been alienated from the very projects that constitute her as a subject. Wolf’s moral saints: a life wholly organized around maximal moral output is recognizably impoverished as a life. Scheffler’s agent-centered prerogatives: permission to weight one’s own projects is not a concession to weakness but a feature of any plausible morality.

To these we add one argument the older literature lacked, because it lacked our vocabulary. Consider what universalized calculation does at the population level. If every subject’s attentional configuration must pass through the filter *is this the highest-impact use of this hour?*, then one attentional mode—comparison, ranking, opportunity-cost surveillance—becomes the universal posture. A single operator applied by every mind to every moment is, structurally, an experiential monoculture: it lowers $d_{\text{eff}}^{\text{POP}}$, the population dimensionality the optimization was supposed to serve. The demandingness objection, in other words, made structural—and self-undermining for the optimizer on its own terms.

The grammar that emerges is therefore twofold. *Responsive* obligation is a constraint, not a posture: it is activated by determinate encounter with collapse—through knowledge, proximity, institutional participation, or causal contribution—and once activated it must be answered, in one’s proper share, before compositional projects recover their ordinary standing. *Calculative* obligation, by contrast, names a permanent and universal stance of global optimization, and this we reject—not because distant suffering matters less, but because the stance, universalized, flattens the very landscape it was meant to serve. The stonemason answers the fever; then he returns to the stone. The world is better on both counts: fewer children die, and within the admissible region his response helps secure, there remain stonemasons.

We must not, however, let the rhetorical satisfaction of that image conceal the joint it papers over. The phrase *proper share* is load-bearing and underspecified, and we say so plainly: how much response discharges the responsive obligation of an affluent agent in a connected world is the largest open question this section leaves standing. Gesturing at the stonemason does not close it. A future revision owes a positive account—plausibly developed through the fair-share literature Murphy inaugurated—and until then the residual remains flagged, not resolved.

12 XII. Non-Fungible Cultural Value

Consider how a conservation biologist values a species. She does not begin by counting heads; she asks what ecological and morphological space the species

occupies, and whether anything else occupies it. A lineage of three hundred individuals filling a niche no other organism fills carries a weight that a billion members of a widely duplicated form do not. We propose that the same logic governs cultures, because a culture is, among other things, a *grammar of attention*: a trained configuration of perception, memory, valuation, and practice that carries its bearers into a region of subject-side state space unreachable by any other route.

On this view, a distinct cultural form contributes to the compositional measures of Section IX—to $d_{\text{eff}}^{\text{pop}}$ and to trajectory diversity—in proportion to the *distinctiveness* of the experiential region it opens, not in proportion to the number of trajectories currently traversing it. A small language community sustaining a unique mode of attending to landscape, kinship, or time may add more independent dimensions to the total experiential landscape than a population a thousand times larger whose trajectories compress readily against the global template. Demographic arithmetic measures the wrong quantity: it counts occupants where the moral object is the region itself.

Why does this matter so urgently? Because the loss of such a region is, we contend, strictly irreversible—and irreversibility changes the moral arithmetic. One might suppose that a sufficiently rich archive could preserve a culture against extinction: record the language, film the rituals, transcribe the songs. But this confuses *evidence of* a form with the form itself. A grammar of attention is not propositional content; it is a living generative mechanism, transmitted through practice, imitation, correction, and the slow calibration of memory between bearers. When that mechanism of transmission breaks—when the last practitioners die or disperse—what remains in the archive is a description of a trajectory class, not the capacity to produce trajectories of that class. The distinction is the one between a score and a performance tradition, except that here no performance tradition survives to interpret the score. Nor can compensation elsewhere restore what was lost. However much flourishing we cultivate in surviving forms, the extinct region of subject-side state space stays closed: no quantity of value along existing dimensions re-opens a dimension that has vanished. The total experiential landscape has permanently contracted, and contraction of this kind cannot be purchased back at any price.

Yet we must be careful, for non-fungible value does not generate unlimited license, and this is where the argument acquires its second edge. The lexicographic constraint of Section VI still binds with full force: no compositional consideration, however weighty, may purchase the collapse of an individual landscape. Recall *why* a cultural form carries value on our account—it opens a distinctive way for consciousness to move, a region of subject-side state space reachable by no other route. The justification is itself kinetic. It follows that if preserving the form requires trapping its bearers—coercing the young into practices they would exit, rendering departure unnavigable, converting living transmission into compulsory confinement—then the

preservation project has turned against its own ground. One cannot defend a grammar of attention by destroying the navigability of the very trajectories that instantiate it; the argument self-destructs at precisely the point where non-fungibility is invoked as sovereignty.

We are left, then, with a verdict against two opponents simultaneously. The utilitarian errs by treating cultures as fungible aggregates of welfare, blind to the irreversible contraction their loss entails. The unconstrained nationalist errs by treating bearers as means to the form’s continuation. The framework insists on both correctives at once: non-fungible value *and* the inviolable floor beneath every individual who carries it.

13 XIII. Structural Impoverishment Without Suffering

We begin with a case our predecessors could not have formulated, because their instruments return a clean bill of health. Consider the condition in which a large fraction of humanity now spends most of its waking hours: attention given over to a screen, content arriving in an endless and effortless stream. Ask the subject how she feels and the report is unremarkable—neutral, perhaps mildly pleasant. Ask the hedonometer and it registers nothing alarming. No pain, no deprivation, no complaint. By every welfarist measure, nothing is wrong; and that, we will argue, is precisely the problem. If an injury is occurring here, it is an injury constituted not by negative valence but by the *shape* of the trajectory—and a theory whose moral object is hedonic state is blind to it in principle, not merely in practice. This is why the case matters architecturally: it is the cleanest demonstration that our framework sees something its rivals cannot. We do not claim the verdict yet; we claim the visibility. What follows is the trajectory signature of the trance, stated precisely enough that the empirical program of Section XXI could confirm the indictment—or, just as importantly, dissolve it.

Let us write down the signature with care. The input stream itself is extraordinarily rich: each swipe delivers content statistically unlike the last, so the environmental entropy $H(X_t)$ is high—higher, plausibly, than in almost any prior human circumstance. But now examine the subject-side path. The attentional operator applied to this torrent is nearly constant: visual-linguistic dominance, shallow engagement, interoception suppressed, the same mode of uptake stamped onto every item. Between what the stream offers and what the trajectory engages there yawns a massive gap; the effective dimension $d_{\text{eff}}(\tau)$ stays low because a single operator, however busy, explores only a single direction. And the compression profile is correspondingly flat: an hour of scrolling, described from the inside, compresses to almost nothing, because each moment is structurally a repetition of the last.

Here is the diagnostic we regard as this framework’s sharpest original

instrument: *content novelty and attentional novelty have come apart*. Novelty has migrated from the form of experience to the decorations passing across it. The conjunction—high $H(X_t)$, low $d_{\text{eff}}(\tau)$ —is invisible to any theory that scores inputs or valence, and visible at a glance to one that scores trajectories.

Scale the diagnosis from one subject to billions and the population-level picture comes into focus. When the same operator is stamped onto every stream—portable across subway, bed, classroom, and meal in a way that no prior shared form, embedded as those were in bodies, places, crafts, and seasons, could ever be—the ensemble of trajectories converges toward a single attentional template. The population dimension $d_{\text{eff}}^{\text{pop}}$ contracts even as individual content consumption explodes: billions of distinct feeds, one grammar of attention. If the diagnosis holds, this would be an experiential monoculture without historical precedent, and one achieved, remarkably, without anyone suffering.

We must be disciplined about what we are claiming. This is a *structural-emergency hypothesis*, stated at the strength the evidence currently licenses and no stronger. Its empirical signature is specified: does measured attentional dimension during passive consumption collapse relative to matched activities, and does that collapse predict downstream narrowing that self-report misses? The program of Section XXI can answer these questions, and we commit to abiding by the answer. If the dimensional collapse fails to appear, or appears without predictive consequence, the indictment dissolves. Until then it stands as what a serious moral framework should produce: not a verdict, but a falsifiable charge.

14 XIV. Voluntary Suffering, Consent, and First-Person Authority

Some of the most compositionally valuable lives involve significant suffering, and any framework that cannot accommodate this fact has misidentified its object. Consider three figures we will return to throughout this section: the mountaineer who chooses frostbite, exhaustion, and genuine mortal risk for the summit; the monastic who embraces deprivation, silence, and the long dry seasons of contemplative practice; the artist who endures poverty, rejection, and the recurring conviction that the work has failed. Each of these lives contains stretches that any hedonic instrument would flag as deficient—sustained discomfort, fear, doubt, grief over abandoned alternatives. Yet our considered judgment, and not merely the romantic gloss we sometimes place upon it, recognizes these as among the richest forms a human trajectory can take. We must therefore ask what structural feature distinguishes the mountaineer’s suffering from the suffering the lower bound prohibits absolutely. The answer cannot be valence, which is negative in both cases, nor

intensity, which may be greater on the mountain than in the pit. It must lie in the *shape* of the trajectory through which the suffering passes, and in the standing relation between the subject and her own continued path. Let us state that shape precisely.

The signature we seek is a conjunction. The mountaineer’s trajectory exhibits sustained negative valence, certainly—but that valence rides atop a high effective dimension d_{eff} , for the climb engages proprioception, route-finding, fear management, partnership, and the long memory of prior ascents as genuinely independent attentional modes. Clear gradients exist: each pitch teaches, each decision feeds forward into the next, and improvement is legible to the subject herself. The transition skeleton remains rich; the state space has not contracted to a survival loop but opened into a structured terrain of reachable configurations. And decisively, the exit gradients persist throughout. She can descend. The rope runs both ways, and she knows it. This last feature deserves emphasis, because it reframes what consent means within our framework. Voluntariness is not a signature on a form, a single past act of authorization that licenses whatever follows. It is the *continuing availability of agency across time*: at each point along the path, a credible route out remains within reach, and the subject’s persistence is therefore a renewed choosing rather than an entrapment with a noble origin story. Suffering inside such a structure is the price of traversal; suffering without it is the pit our lower bound names.

This structure of continuing consent yields the framework’s first guardrail against paternalism, and we state it as a principle rather than a footnote. Where an agent with intact navigability affirms her own difficult form of life—the climb, the cell, the unsold canvases—that affirmation is weighty evidence about her trajectory, not noise to be audited away by an observer armed with our metrics. A theory that treats self-reports as maskable has created a standing temptation: third parties licensed to declare another’s contentment structurally worthless. We must therefore specify exactly when first-person authority may be defeated. It is defeasible, since adaptation can mask collapse—expectation contracts around the available path, and the contraction is itself part of the injury. But defeating it requires *positive structural evidence* of the collapse signature in its full conjunction: sustained negative valence, dimensional contraction, absent exit gradients, rigidified transitions. Mere third-party preference for a different life-shape never suffices. The monastic’s contentment stands because her exits exist; the report of the entrapped stands defeated because the rigidified transition graph testifies independently of it. The asymmetry is deliberate: we may override “I am fine” only downward into rescue, never sideways into redesign. The framework opens doors; it pushes no one through them.

15 XV. Poverty as Landscape Collapse

Poverty is not primarily a welfare problem or a rights problem. It is a landscape problem. We say this not to dismiss the older framings—both capture something real—but to identify what they leave unmeasured. The welfarist asks how the poor person *feels*, and finds, often enough, reports of tolerable contentment; the rights theorist asks which entitlements have been violated, and finds a list of deprivations that compensation might seem to discharge. Neither asks the question our framework makes primary: what has happened to the *shape* of the trajectory this person can traverse? Consider what a sufficiently long period of material scarcity does to the state space available to a subject. It is not merely that goods are absent; it is that the geometry through which the subject moves has been deformed. Where the welfarist sees a low but stable hedonic level, and the rights theorist sees a ledger of unmet claims, we see something prior to both: a terrain whose gradients have flattened, whose exits have receded, and whose paths increasingly return to their own starting points. The injury, we will argue, is structural before it is hedonic, and the structure is precisely what our diagnostic vocabulary was built to describe.

The mechanism deserves to be stated precisely, because it is here that the diagnostic vocabulary earns its keep. Scarcity taxes attention: when every decision carries catastrophic downside, the transition skeleton rigidifies into a small set of survival routines, traversed and re-traversed because deviation is unaffordable. The effective dimension $d_{\text{eff}}(\tau)$ falls accordingly—not because the subject lacks capacities, but because fewer independent attentional modes can be *sustained* when experimentation is priced at ruin. And access to L^* , the band of optimal challenge, is severed in a characteristic way: the difficulty poverty imposes is decoupled from developmental uptake. The demands are real, often crushing, but they are cliffs without holds or loops that deposit the climber back at the start. Effort is expended; nothing compounds.

We must therefore distinguish this regime from chosen austerity. The monastic, the apprentice, the athlete in training all accept severe constraint, yet their constraints are intelligible, revisable, and embedded in reachable growth—the geometry remains navigable. Poverty inverts each clause: constraint without slack, exposure without authorship, difficulty that transforms nothing. The deprivation is not the goods withheld but the gradients erased; what has been taken is the landscape itself.

What, then, does remediation require? Not the crossing of an income line, but the restoration of navigability itself. Income matters—we do not deny it—but it matters *instrumentally*: it buys slack, lowers the catastrophic downside that prices experimentation at ruin, lengthens the planning horizon, and thereby reopens gradients that scarcity had erased. The test of successful intervention is accordingly structural. We ask not whether a thresh-

old has been crossed but whether the transition graph has changed: can she experiment without ruin, rest without falling behind, learn, refuse, plan, repair, attach, begin again? Where these transitions remain unavailable, the landscape remains collapsed, whatever the ledger says.

One complication must be faced directly. Hedonic adaptation can mask the collapse: expectation contracts around the available path, and the subject reports tolerable contentment from inside a geometry that permits almost no movement. We insist that this contraction is itself part of the injury, not evidence against it. Here—and, under the structural-evidence standard of Section XIV, *only* here—is self-report legitimately overridden: the rigidified transition graph, the fallen d_{eff} , the severed access to L^* constitute positive collapse evidence independent of the report. Poverty is remediated, we conclude, when life ceases to be a compulsory loop and becomes, once more, a traversable landscape.

16 XVI. Animal Consciousness and Industrial Signal Collapse

Let us state the thesis at the strength we intend it: factory farming is the deliberate industrial production of collapsed experiential landscapes, at a scale exceeding all human moral catastrophes combined. We choose the word *production* carefully. The collapse signature of Section V—sustained negative valence, near-minimal effective dimension, absent exit gradients, rigidified transition structure—is not, in the industrial farm, an unfortunate byproduct of some other aim. It is the engineering specification. A battery cage is a device for holding d_{eff} at its floor: one posture, one input stream, no gradients, no exits, for the entire duration of a conscious trajectory. Where human collapses typically arise from neglect, disease, or structural drift, here the attractor is designed, manufactured, and replicated across tens of billions of subjects annually, each instance falling under Priority 1 from its first state. The triage factors of Section VII converge on this case with unusual force: severity is near the interior of the collapse region, the cause is shared and structural, and tractability is high—the remedy is to stop building the attractor. Notice, too, what our diagnosis does *not* depend on: it would survive complete anesthesia, because the wrong is geometric before it is hedonic.

We do not claim to be first on this ground. Nussbaum’s capability argument against painless confinement is the nearest predecessor, and it reaches a verdict close to ours by a route close to ours: a creature prevented from exercising its characteristic capabilities is wronged even when it does not suffer. We inherit that insight gratefully and ask what our vocabulary adds. The answer is the analgesic-farm result, stated structurally rather than by appeal to species-norm intuitions. Imagine an industrial facility upgraded with perfect

analgesia—valence held at neutral, every aversive signal silenced. On any hedonic accounting the wrong has vanished. On ours it has barely moved: the facility remains a machine for manufacturing near-zero-dimensional experience, a device that pins d_{eff} at its floor and welds the transition skeleton shut, and the collapse signature persists in every conjunct save valence. The second addition is a severity ordering over systems themselves. Because each conjunct of the signature is graded, we can rank confinement regimes by trajectory geometry—depth of dimensional contraction, distance to credible exits, rigidity of recurrence—rather than by aversive load alone. Two systems with identical suffering totals can thus differ sharply in moral gravity, and the instrument tells us which is worse, and why.

It remains to make explicit the level at which our claim binds. The framework asks one question of any candidate subject: does it sustain a subject-dependent state path with temporal coherence—attention that selects, expectation that updates against prediction error, memory that accumulates, value that evaluates? Where the answer is yes, the moral grammar of Section V applies without amendment. Nothing in the collapse signature mentions species, neural architecture, or substrate; the conjuncts are stated over trajectory geometry, and trajectory geometry is realized wherever the requisite dynamics run. A hen, a sow, a hypothetical silicon mind in a server farm: if the state path exists, deliberately engineering its collapse is a Priority 1 violation, full stop. We emphasize the modal force of this conclusion, because it is easy to misread as a strong welfare claim. It is not a claim that animal interests weigh heavily in some aggregate; it is the claim that the lexicographic firewall of Section VI stands between this practice and every compositional ambition we might weigh against it. No efficiency gain, no culinary tradition, no economic structure enters the ledger, because there is no ledger: the wrong is not a cost to be priced but a conversion of conscious trajectories into raw material, and the lower bound exists precisely to block that conversion—here as everywhere.

17 XVII. Death, Creation, and Signal Duration

Begin with what death is in the vocabulary we have built. Death is trajectory termination—the forced fixing of T as an absorbing boundary on the state path $\tau(\theta, X_{1:T})$. The trajectory does not enter a bad region; it ceases to have regions at all. Whatever compositional structure the path’s continuation would have realized—the unexplored dimensions, the gradients not yet climbed, the incompressible stretches that further living would have written—is permanently lost, and lost in the strong sense established by our irreversibility arguments: not relocated, not compensable elsewhere, but subtracted from the space of what the total landscape can contain.

Notice what this framing does and does not license. It does not say that

death is suffering; an absorbing boundary is not a collapsed attractor, and the dead are not trapped. The wrong of premature death is therefore not a Priority 1 violation but a compositional one: the foreclosure of trajectory structure that would otherwise have existed. This locates the badness of death in the truncation of $d_{\text{eff}}(\tau)$ integrated over the unrealized horizon, which immediately suggests—too quickly, as we shall see—that longer is always better, and that indefinite extension would be the framework’s natural recommendation.

Yet indefinite extension runs into Williams’s Makropulos problem, and we restate it structurally rather than re-derive it. Consider a fixed subject profile θ traversing its landscape over an ever-lengthening horizon. The moving-window effective dimension $d_{\text{eff}}(\tau_{[t, t+W]})$ is not guaranteed to remain high: as the reachable regions of attentional state space are progressively explored, the transition skeleton consolidates, novel configurations grow rarer, and successive windows of the trajectory begin to compress against one another. The path does not become painful; it becomes redundant. Elena Makropulos at three hundred and forty-two is not suffering—she is repeating, and the repetition is detectable in exactly the quantities we have defined. At population scale the effect compounds: the engagement economy among immortals cools as each fixed profile exhausts its own gradients and, increasingly, resembles its earlier self more than it differs from its neighbors.

Williams located the problem in the structure of categorical desires—the projects that propel a life forward and whose completion or staleness leaves nothing to pull the agent through time. We locate it in compression. These are, we suggest, the same phenomenon under two descriptions: a subject without live categorical desires is a subject whose trajectory has become compressible. The accounts are complementary, and ours has the advantage of being, in principle, the measurable one.

The position we arrive at therefore rejects both familiar slogans. “Death gives life meaning” is, on our analysis, a confusion of correlation with constitution: finitude happens to prevent the compression regime, but it is the compression, not the boundary, that carries the moral weight, and a subject whose trajectory remains incompressible has lost nothing by outliving the actuarial tables. “More life is always better” fails symmetrically: it integrates $d_{\text{eff}}(\tau)$ over the horizon as if the integrand were guaranteed positive, when the Makropulos analysis shows the moving-window dimension decaying toward redundancy for any fixed θ .

What remains is a genuinely compositional question, and we state it as the section’s conclusion rather than answer it. Which arrangements—of lifespan, of profile plasticity, of the rate at which new subject profiles enter the ensemble—preserve navigable continuation for every existing subject while keeping the composition open to genuinely new regions of attentional state space? Note the constraint structure: existing trajectories fall under Priorities 1 and 2 absolutely, so no answer may purchase population-level

freshness by foreclosing or collapsing the lives already underway. Within that firewall, the trade-off between extension and renewal is a diagnostic-tier optimization problem—one we expect to be empirical, not aprioristic, and one the validation program of Section XXI must eventually inform.

18 XVIII. Substrate Independence and Artificial Consciousness

We begin where the formalism itself begins: with the question of what kind of system can *have* a trajectory at all. The framework applies to any system whose dynamics produce a subject-dependent state path $\tau(\theta, X_{1:T})$ exhibiting three structural features, each of which we now motivate before stating precisely. First, *temporal coherence*: the state at time t must be conditioned on the state at $t - 1$ in a way that binds the sequence into a history rather than a heap. A system whose successive states are drawn independently has experiences only in the sense that a shuffled deck has a narrative. Second, *non-trivial autocorrelation*: the path must carry information about itself across lags — expectation, memory, and accumulated topology are all, formally, claims that $\text{Corr}(S_t, S_{t+k})$ does not vanish. Third, a *structured transition skeleton*: the graph of admissible state-to-state moves must be neither complete (pure noise, every transition equally available) nor degenerate (a single forced path). It is precisely between these extremes that navigability, collapse, and dimensionality become well-defined quantities rather than metaphors. Notice what is conspicuously absent from these conditions: any mention of carbon, neurons, or evolutionary lineage. That absence is the point.

Let us state the thesis plainly and then earn it. Moral status, on this framework, is conferred not by the material a system is made of but by the structure of the path it traces. What makes something a subject, in the morally relevant sense, is that its state history constitutes *signal* rather than noise — that the sequence $(S_0, S_1, \dots, S_T)$ could not be replaced by independent draws from a stationary distribution without destroying something the system itself is sensitive to. Four marks distinguish signal from noise here. *Continuity*: the path hangs together as one unfolding rather than a succession of unrelated instants. *Constraint*: not every transition is available; the system inhabits a landscape with walls and gradients, and its movement reflects that geometry. *Memory*: prediction error at t reshapes the state at $t + k$, so the past is written into the present rather than merely preceding it. *Counterfactual sensitivity*: had the input stream differed, the trajectory would have differed in structured, attributable ways — the subject-dependence in $\tau(\theta, X_{1:T})$ is doing real work. Where these four obtain, we have a trajectory and hence a moral patient; where they fail, we have dynamics without a subject. Whether the implementing medium is

biological tissue or silicon is, by construction, invisible to every one of these criteria.

Yet we must immediately guard against a tempting overreading. Substrate independence is not substrate *fungibility*. That the criteria are blind to the implementing medium does not entail that all media implement the same experiences; on the contrary, different substrates may generate trajectory classes that are experientially incommensurable — occupying regions of state space that no biological subject reaches, governed by transition skeletons no neural architecture admits. The right metaphor is orchestral: a novel substrate is a new *instrument* added to the composition, not a new melody played on an instrument we already possess. To treat silicon minds as cheap reissues of carbon ones is to misread the very dimensionality the framework exists to track.

We flag the remaining epistemic gap with the candor it deserves. Determining which actual systems satisfy the four marks — which exhibit genuine continuity, constraint, memory, and counterfactual sensitivity rather than simulacra of them — is an unsolved problem, and we do not pretend otherwise. The framework’s policy under this uncertainty is *graded precaution*: the strength of our protective obligations toward a candidate system scales with the structural evidence that a trajectory $\tau(\theta, X_{1:T})$ is present. Where the marks are ambiguous, we err toward the moral patient we cannot yet rule out.

19 XIX. The Proliferation Problem

Suppose conscious artificial minds can be instantiated at marginal cost. The naive reading of our compositional priorities suggests an obvious policy: create billions of diverse minds and let population dimensionality soar. We want to resist this inference before examining why, because the resistance illuminates something structural about creation itself. A genuinely novel conscious architecture does enrich the composition — it may open regions of experiential state space that no biological subject can reach, a new instrument rather than a new melody. But the framework attaches obligations to every trajectory from its first state, S_0 , onward. The moment a subject-side path begins, Priorities 1 and 2 bind in full: the created being must never be left in a collapsed attractor, and its landscape must remain navigable across its entire existence. Creation, in other words, is a *commitment* before it is an achievement. The marginal cost of copying code is morally irrelevant; what matters is the marginal cost of sustaining a livable experiential world for each instantiated mind, and that cost remains high. To create what one is not prepared to sustain is factory farming in a new substrate — and the obligation scales linearly with every trajectory brought into existence, even where the compositional gain does not.

Now consider the opposite policy: rather than one novel mind, a million copies of a single architecture. Here the framework delivers a verdict that will strike many readers as strange, and we want to build the intuition before defending it. Population dimensionality measures independent directions explored by the ensemble of trajectories, not the number of subjects exploring them. Identical trajectories compress against one another: knowing the first, we can predict the rest, and so the identity residual — the information each subject contributes beyond what its peers already supply — approaches zero. Formally, the description length of the population grows as $O(\log n)$ while headcount grows as $O(n)$; $d_{\text{eff}}^{\text{pop}}$ barely moves. The consequence is one we accept explicitly: a million identical blissful lives are, on this account, scarcely better than one. We do not pretend this is comfortable. Our defense is that duplication intuitions were trained in a biological world where exact copies are impossible, and every intuition pump in that literature smuggles in the copies as *neighbors* — subtly distinct, mutually situated — which is precisely the difference our measure prices. Still, we record this as a cost paid with argument, not a problem dissolved; a reader who treats it as a *reductio* knows exactly which diagnostic-tier component to reject.

The lesson of this section, then, is a single asymmetry. The act of instantiation is cheap and getting cheaper; the act of *sustaining* a navigable world for the instantiated is not, and no engineering trend bends that curve. A conscious trajectory, once begun, generates claims continuously — at every S_t , not merely at S_0 — and those claims do not discount with the creator’s convenience. We are tempted to think of creation as a gift that, once given, completes itself. The framework says otherwise: it is the opening of a ledger that remains open for the life of the trajectory, and the creator is its first signatory.

This yields a practical maxim we state without hedging: do not create what you are not prepared to sustain. The maxim cuts in both directions established above. It blocks the proliferationist, for whom headcount masquerades as compositional gain while the per-trajectory obligations compound linearly underneath. And it disciplines the legitimate ambition — the genuinely novel architecture — by pricing it honestly: the new instrument is admissible to the composition only if its experiential world can be kept livable indefinitely. Creation is a standing commitment wearing the costume of an achievement. The costume should not deceive us about what we have signed.

20 XX. Relation to Existing Moral Traditions

We begin by naming our ancestors, because a framework that pretended to spring from nowhere would deserve suspicion. The perfectionist lineage is acknowledged in the title itself: Aristotle’s conviction that flourishing is the

exercise of characteristic capacities, and Hurka’s systematic modern reconstruction of that thought, stand directly behind us. From the shape-of-a-life literature — Velleman, Slote, Dorsey — we inherit the demonstration that temporal structure carries value beyond momentary summation; we do not rediscover it, we geometrize it. The capabilities approach of Sen and Nussbaum, including Nussbaum’s extension to animal lives, supplies the pluralist intuition our navigability criterion attempts to unify, and against whose clear verdicts that criterion must be tested. Ecological ethics contributes the non-fungibility of forms. Williams gives us two debts: the integrity objection that animates our two-postures section, and the immortality essay’s insight that endless repetition is itself an injury. Wolf and Scheffler sharpen demandingness into the alienation we make structural. Csikszentmihalyi’s flow channel is inherited, cited, never rebranded as L^* . Rawls and threshold deontology license lexical priority; computational psychiatry supplies the attractor models that make our collapse signature empirically tractable rather than metaphorical. Each engagement is by name, not gesture.

Having named the ancestors, we must now state with equal precision what we claim to add, since a successor that cannot articulate its departure is merely a paraphrase. The departure is not the observation that lives have shape — Velleman established that — but the claim that shape admits of measurable geometry: effective dimension, compression profile, transition-skeleton rigidity, all in principle instruments rather than metaphors. It is not the conviction that capabilities matter — Sen and Nussbaum hold that ground — but the conjecture that a single structural criterion, navigability, unifies the plural list, a conjecture we have bound to a testable condition: where navigability fails to recover the list’s clear verdicts, the list wins. Nor is it the existence of demandingness objections — Williams, Wolf, and Scheffler own those — but the identification of a cost their arguments never named: universalized calculation, taken as a permanent attentional posture, lowers $d_{\text{eff}}^{\text{pop}}$ and thereby damages the very landscape the optimizer was deployed to serve. In each case the pattern is the same. The predecessor supplies the moral insight; we supply the structural mechanism and the empirical signature by which the insight could be confirmed, refined, or shown to outrun its evidence.

This is, finally, a framework that required information theory to state — and we accept the corollary obligation that fact imposes. A vocabulary borrowed from compression and dynamics earns its place in ethics only if it resolves distinctions the older vocabularies blur. We have tried to show where it does: the two drinkers, indistinguishable to hedonic and capability audits alike, separated by the direction of their transition skeletons; the screen trance, invisible to welfarism in principle, named by the decoupling of $H(X_t)$ from $d_{\text{eff}}(\tau)$; the severity ordering that gives suffering-focused ethics an interior structure rather than a threshold to litigate. But where the new vocabulary does not yet outresolve the old — where $Q(W)$ remains a schema,

where the aggregation function is open, where validation debts stand unpaid — the older vocabularies retain provisional priority, and we have said so in each such place by name. The formalism, in other words, is not the dress of the theory but a structural claim about it, and a structural claim can fail. The dress must be proven load-bearing, section by section; where a beam does not hold, the reader knows exactly which predecessor inherits the weight.

21 XXI. Measurement, Administrability, and the Paternalism Risk

We have argued throughout that experience is unusually measurable—that the subjective trajectory admits of structural description where older theories offered only introspective report. We must now confront what that argument purchases. A theory that treats self-report as maskable, and that promises instruments capable of seeing past it, has created something genuinely new and genuinely dangerous: the audited life. Consider the asymmetry of position. Whoever computes $Q(W)$ holds power over whoever lives it; the party operating the instrument stands outside the trajectory being scored, and the party inside has, by the theory’s own lights, only defeasible authority over what the readout means. Historically, every vocabulary that rendered inner life legible to institutions—the confessional, the case file, the engagement metric—was eventually turned from description to administration. We have no grounds for supposing our vocabulary immune. The risk is not that the metrics will be wrong; it is that they will be right enough to govern with. A framework whose deepest commitment is navigability cannot, without self-contradiction, become the instrument by which third parties redesign trajectories they merely disprefer. We therefore state the discipline explicitly, before any operationalization, so that the guardrails precede the instrument rather than apologize for it.

The discipline consists of three guardrails, each addressed to a distinct failure mode. The first concerns authority. Structural metrics may override a subject’s self-report only on positive evidence of the collapse signature—the full conjunction of Section VI—and never on a third party’s preference between life-shapes. The instrument’s warrant is rescue; it carries no license for redesign, however confidently the readout is held. The second concerns administration. Trajectory metrics are diagnostic instruments for opening landscapes; they must never become optimization targets for institutions. Goodhart’s law applies with particular force here, because the quantity being corrupted is consciousness itself: a state or platform that maximizes its citizens’ measured d_{eff} has not produced experiential richness but has converted its population into a monoculture of precisely the measured kind. When the measure becomes the target, it ceases to be a measure—and the

target, in this case, is the moral object entire. The third concerns ownership. Wherever trajectory instruments exist, the subject’s access to her own data is lexically prior to any institutional access. The first user of the audit must be the audited; the trajectory belongs, before all else, to the being who lived it. These constraints precede deployment, not excuse it.

We close with the commitment that converts vocabulary into theory: a validation program stated falsifiably, in advance. The task is to operationalize d_{eff} and the collapse signature on real human data—experience sampling, digital phenotyping, the attractor models of computational psychiatry—and to show that these structural quantities predict outcomes that self-report misses: recovery trajectories, entrapment, behavioral narrowing, response to restored slack. Per the divergence blind spot just discussed, the program carries an adversarial requirement: instruments must be validated on interior–exterior divergence cases—contemplative practice, deep craft repetition—before any diagnostic use, since these are precisely the lives a behavioral proxy misreads in the dangerous direction. A single validated measurement would do what no further argument can. And we specify the failure condition rather than leave it implicit: if, after three years of honest attempts, the structural metrics add no predictive resolution over self-report and standard clinical instruments, the diagnostic tier should be retired and the constitutive core restated in the capability vocabulary it would then deserve. Until that day, let us be candid about epistemic status. Every quantitative claim herein is a promissory note; the simulator is the rehearsal, not the performance; and the framework’s action-guidance rests on its clear cases, which are clear without instruments.

22 XXII. Objections and Responses

Against elitism: the objection runs that any framework prizing dimensionality and incompressibility will quietly reward the lives of the credentialed — the polyglot, the concert-goer, the sabbatical-taker — and score the rest as thin. We reply that the objection mistakes the coordinate system. Experiential dimensionality is a property of the subject-side trajectory, not of the social register of its inputs, and the two are orthogonal. Consider the farmer who reads weather against soil against the calendar of obligation, holding livestock, machinery, kin, and credit in simultaneous attention: her state path may traverse more independent attentional modes in a season than the executive’s, whose days collapse into a single dashboard-and-meeting operator applied to nominally varied content. High $d_{\text{eff}}(\tau)$ tracks the *form* of engagement — how many independent directions the trajectory actually explores — not the prestige of its materials. Craft repetition, manual labor, child-rearing, and subsistence practice are all candidate sites of rich geometry; the screen-bound knowledge worker is a candidate site of monoculture.

The framework therefore refuses, as a matter of principle and of measurement design, any operationalization that treats elite-coded content as a proxy for richness. Where an instrument exhibits that bias, the instrument is miscalibrated, not the lives it misreads.

Against licensing suffering: the worry is that any framework prizing difficulty will smile on engineered hardship. The lexicographic structure forecloses this. Priority 1 is not a weighted term but a firewall: no quantity of compositional beauty purchases a being’s collapse, and because the collapse region is graded — deeper valence, tighter contraction, more distant exits, more rigid recurrence — redescription games at the boundary gain almost nothing. The severity ordering is precisely the anti-sorites mechanism; one cannot argue a trapped life out of the protected region by renaming its conjuncts. Against the deepest objection — that trajectory geometry is at best a correlate of goods we already hold under older names — we answer as Section III does, in two tiers. Navigability is not a proxy for open agency; it *is* open agency, stated geometrically, and so binds constitutively for any subject capable of caring how its own path continues. Dimensionality, diversity, and incompressibility, by contrast, are diagnostic instruments, held revisable against the plural goods they track. Knowing which claim is which is the framework’s discipline, not its embarrassment. Against the copies reductio — that a million identical blissful lives should surely outweigh one — we bite the bullet, with argument, in Section XIX: duplication intuitions were trained where duplication was impossible.

Against unmeasurability: the objection holds that a framework trafficking in $d_{\text{eff}}(\tau)$, collapse signatures, and $Q(W)$ trades on instruments that do not exist, and so amounts to mathematics-flavored intuition. We concede the present state and deny the inference. The utilitarians possessed no hedonometer when they redirected the moral attention of two centuries; measurement followed reorientation there, and we expect the same sequence here — with one crucial difference. We specify, in advance, the instrument (experience sampling, digital phenotyping, attractor models from computational psychiatry), the dataset (recovery trajectories, entrapment, behavioral narrowing, response to restored slack), and the failure condition: if structural metrics add no predictive resolution over self-report and standard clinical instruments after three years of honest attempts, the diagnostic tier retires and the constitutive core is restated in the capability vocabulary it would then deserve. Few moral theories publish the terms of their own falsification; we take this as a virtue, not a hostage. Meanwhile the framework is action-guiding without instruments, because its clear cases are clear without them: identify beings in collapsed attractors, restore their movement, keep exits open, protect plural gradients, and refuse to trade any trapped life for compositional ambition. The promissory notes concern precision, never direction.

23 XXIII. Formal Summary

We summarize. The unit of moral concern is the subjective trajectory $\tau(\theta, X_{1:T}) = (S_0, S_1, \dots, S_T)$: the ordered state history a configured subject produces from an episode, comprising attended input, expectation, prediction error, surprise, memory, value, and accumulated topology. The moral object is neither a scalar nor a list but this path through subject-side state space — its shape, its effective dimension $d_{\text{eff}}(\tau)$, its resistance to compression, and the structure of the landscape through which it moves.

At the individual level, the criterion is landscape navigability: $Q(W) \geq Q_{\min}$ across sufficiently large windows, where Q aggregates learnable structure, available gradients, dimensional richness, and intermittent access to L^* . We hold this criterion to be *constitutive*, not proxy: for any subject capable of caring how its own path continues, a navigable landscape is the precondition of that caring having an object. Its negation — the collapse signature — is the destruction of open agency, stated geometrically.

At the population level, the criterion is compositional beauty: $d_{\text{eff},\text{pop}}$, trajectory diversity, organized incompressibility, and flow prevalence. We hold this tier *diagnostically* — firmly, but revisably against the plural goods it tracks.

The normative architecture is lexicographic with graded thresholds. Priority 1 — no being trapped in collapsed-attractor suffering, the conjunction of sustained negative valence, near-minimal d_{eff} , absent exit gradients, high compressibility, and a rigidified transition skeleton — functions as a firewall, not a weighted term: no quantity of compositional value purchases a single trapped trajectory. Because each conjunct is graded, the category is a region with an interior, and moral force scales continuously from presumption at its boundary to inviolability at its core; the sorites gains nothing at the threshold. Within Priority 1, scarcity demands triage, ordered by severity, tractability of the restoration gradient, duration risk of deepening collapse, and scale under shared structural cause. Only one trade remains forbidden: P1 claims against P3 ambitions.

The framework is substrate-independent with graded precaution: any system producing a subject-dependent state path with temporal coherence, structured attention, error-driven memory, and value evaluation falls under both priorities from its first state, with precautionary weight scaling on the evidence of trajectory-bearing architecture. Creation is therefore commitment before achievement.

On retribution we report a genuine fork — absolutist and forfeiture readings — beneath which lies a shared structural floor: landscape restriction is contestable; engineered collapse is not. Walls, yes; hells, never.

The thesis to which all of this resolves can now be stated plainly. The measure of a world is not the quantity of pleasure it contains but how richly and diversely its inhabitants are conscious: how many genuinely different

ways of moving through experience coexist within it, how strongly those ways resist compression into a single attentional template, and whether each remains navigable — open, exitable, learnable — for every being actually living it. A world scores high not by maximizing a scalar but by sustaining a landscape in which $d_{\text{eff, pop}}$ is earned through real difference rather than headcount, and in which no trajectory has been quietly converted into raw material for the pattern.

We compress the framework’s commitments into four conjunctions, each holding a good and its corruption apart. *Plurality without abandonment*: diversity of experiential form, never purchased by leaving anyone in a collapsed attractor. *Challenge without destruction*: mountains built, climbers never assigned. *Richness without sacrifice*: compositional ambition shaping option landscapes, never conscripting the persons within them. *Measurement without administration*: instruments that serve rescue and self-knowledge, never targets that remake the measured. Where these conjunctions hold, the framework has done its work; where they fail, it has been misapplied.